

Week 1 Syllabus â€” Hardware Manufacturing Fundamentals **Course:** 5-Year Doctorate-Level Hardware Manufacturing Program **Target:** RK3588 carrier board + custom case + immutable faeOS image **Schedule:** 5 days/week ã— 3 hrs/day (Theory 1.5h, Lab 1h, Outdoor/Study 0.5h) **Budget:** â‚¬1,000 tools (phased); all learning via free resources --- ## Day 1 (Monday) â€” Course Orientation & Semiconductor Physics Refresher ### Theory (1.5h) - **Course structure & goals:** 300 credits / 260 weeks / 5 product lines shipped - **Hardware target:** RK3588 SoC (ARM Cortex-A76ã—4 + A55ã—4, Mali-G610, NPU 6 TOPS) - **Semiconductor basics:** Si crystal structure, doping (n/p-type), bandgap, Fermi level - **PN junction:** Depletion region, forward/reverse bias, I-V curve, diode equation - **MOSFET physics:** MOS capacitor, threshold voltage, inversion layer, I_D - V_{GS}/V_{DS} **Free Resources:** - *Semiconductor Physics* â€” MIT 6.007 OpenCourseWare (lecture notes, problem sets) - *MOSFET Operation* â€” All About Circuits textbook (vol. 3, ch. 6) - *RK3588 Datasheet* â€” Rockchip official (public, 200+ pages) ### Lab (1h) - **LTSpice tutorial:** Install, schematic capture, DC sweep of diode & MOSFET - Simulate: 1N4148 I-V curve, 2N7000 $V_{GS(th)}$ extraction, CMOS inverter VTC - Export plots â†’ `week1/day1\_spice/` for portfolio ### Outdoor/Study (0.5h) - Read: *The Transistor* (IEEE Milestone article, free PDF) - Watch: Ben Eater "How Transistors Work" (YouTube, 15 min) --- ## Day 2 (Tuesday) â€” PCB Design Fundamentals & KiCad Mastery ### Theory (1.5h) - **PCB stackup:** FR-4 properties, copper weight, dielectric constant, impedance control - **Trace geometry:** Width/space for 50Î© single-ended, 90Î©/100Î© differential - **Via types:** Through, blind, buried, microvia; aspect ratio limits - **Design rules:** IPC-2221 (free summary), current capacity, thermal relief - **KiCad 8 workflow:** Schematic â†’ Footprint â†’ PCB â†’ DRC â†’ Gerber/Drill/Assembly **Free Resources:** - *KiCad 8 Official Tutorial* (kicad.org/doc) - *IPC-2221 Design Guide Summary* â€” Sierra Circuits / Eurocircuits free whitepapers - *High-Speed PCB Design* â€” Rick Hartley notes (free PDF via Altium/SI-list archives) - *RK3588 PCB Layout Guide* â€” Rockchip application note (request via distributor or GitHub mirrors) ### Lab (1h) - **KiCad project:** Create `rk3588\_carrier` project - Draw: Power input (USB-C PD 5V/3A), buck converter (MP2359), LDO (AP2112 3.3V/1.8V) - Place: RK3588 BGA footprint (19ã—19mm, 0.65mm pitch, 361 balls â€” use ultra-librarian free symbol) - Route: DDR4 x32 (2ã—16-bit) â€” length match $\hat{A}\pm 50ps$, 40Î© single-ended - Run DRC â†’ export Gerbers â†’ `week1/day2\_kicad/` ### Outdoor/Study (0.5h) - Read: *Right the First Time* (Lee Ritchey, Ch. 1-2 free excerpts) - Browse: JLCPCB / PCBWay capability pages (track min 0.1mm, via 0.2mm, 4-layer â‚¬2/5pcs) --- ## Day 3 (Wednesday) â€” Power Electronics & RK3588 Power Tree ### Theory (1.5h) - **RK3588 power domains:** VDD_CPU_L (0.85V), VDD_CPU_B (0.85V), VDD_GPU (0.85V), VDD_NPU (0.85V), VDD_LOGIC (0.85V), VCC_1V8, VCC_3V3, VCC_5V0, DDR_VDDQ (1.1V/1.2V) - **PMIC:** RK806-1 (IÂ²C, 8

bucks + 6 LDOs) â€” datasheet & register map - **Buck converter design**: Synchronous vs. non-sync, inductor selection, output ripple, compensation - **PD controller**: CH32X035 / IP2721 for USB-C 5V/3A negotiation - **Power sequencing**: PMIC enable â†’ domain ramps â†’ POR â†’ PMU firmware **Free Resources**: - **RK806-1 Datasheet** â€” Rockchip (public via GitHub mirrors) - **Switching Regulator Design** â€” TI "Power Supply Design Seminar" slides (free) - **Understanding Buck Converters** â€” Analog Devices AN1197 (free) - **USB-C PD Spec** â€” USB-IF free download (register for free account) ### Lab (1h) - **LTSpice**: Design 5Vâ†’0.85V/4A buck (MP2359) â€” choose L, C_{out}, compensation - Simulate: Load step 0â†’4A, measure overshoot/undershoot, phase margin - **KiCad**: Add PMIC schematic page, wire IÂ²C to RK3588, add decoupling (0201 0.1ÂµF Ã— 4 per ball group) - Export BOM â†’ `week1/day3\_power/` ### Outdoor/Study (0.5h) - Read: **PMIC Layout Guidelines** â€” RK806-1 app note (Rockchip GitHub) - Calculate: Total power budget (RK3588 max ~12W, carrier board budget 15W) --- ## Day 4 (Thursday) â€” DDR4 Memory Interface & Signal Integrity ### Theory (1.5h) - **DDR4 basics**: Bank groups, burst length 8, preamble, write leveling, read/write training - **RK3588 DDR PHY**: 2Ã—16-bit channels, up to 3200 MT/s, ODT, V_{REF} training - **Signal integrity**: Eye diagrams, ISI, crosstalk, SSO noise, termination (ODT vs. fly-by) - **PCB routing**: T-topology vs. fly-by, length matching (byte lane Â±10ps, CK Â±5ps), layer assignment - **Simulation**: IBIS models, HyperLynx/SIWave (free trials) or open-source (QUCS-S) **Free Resources**: - **DDR4 Design Guide** â€” Micron TN-47-01 (free PDF) - **RK3588 DDR Layout Guide** â€” Rockchip app note - **Signal Integrity Primer** â€” Eric Bogatin free videos (YouTube) / "Signal and Power Integrity" Ch. 1 free - **IBIS Models** â€” Micron/Samsung/SK Hynix free download ### Lab (1h) - **KiCad**: Route DDR4 Channel A (16 bits + DQS/DQS# + CK/CK# + CKE + CS + ODT) - Constraints: 50Î© SE, 100Î© Diff, length match DQâ†’DQS Â±5mm, CKâ†’DQS Â±2mm - Place: 0402 series termination (22Î©) near RK3588, decoupling at each V_{DDQ} ball - **QUCS-S**: Import IBIS, simulate DDR4 read/write eye at 2400 MT/s - Export: Constraint manager rules â†’ `week1/day4\_ddr/` ### Outdoor/Study (0.5h) - Read: **High-Speed Digital Design** (Johnson & Graham) Ch. 3 free excerpt â€” transmission line basics - Watch: Robert Feranec "DDR4 Layout" (YouTube, 30 min) --- ## Day 5 (Friday) â€” Bring-up Strategy & Tooling Plan ### Theory (1.5h) - **Bring-up phases**: 1. Power-on (no SoC) â€” check rails, PMIC IÂ²C 2. SoC soldered â€” JTAG/SWD, UART0 boot ROM 3. DDR training â€” RK3588 DDR PHY firmware (BL31/ATF) 4. Storage â€” eMMC / SD / NVMe 5. Display â€” MIPI DSI / HDMI / DP 6. Peripherals â€” USB, PCIe, Ethernet, Audio - **Debug interfaces**: JTAG (ARM DSTREAM compatible), SWD, UART (1.8V), trace (ETM/ETB) - **Test equipment roadmap** (phased): - Month 1-3: USB microscope (â„¢50), TS100 iron (â„¢60), DMM (â„¢30), practice kits (â„¢50) = **â„¢190** - Month 3-6: Hot-air (â„¢110 shared), 4-ch scope (â„¢380 shared/

makerspace) = **â,¬0 new** - Month 6+: VNA (â,¬200 kit), logic analyzer (â,¬40), thermal cam (â,¬200 rental) - **Version control**: Git for KiCad (diff via `kicad-cli sch/bom`), Gerber review checklist **Free Resources**: *RK3588 Boot Flow* â€” Rockchip SDK docs (GitHub: rockchip-linux/rkbin) - *ARM Trusted Firmware* â€” Porting guide (free) - *Buildroot / Yocto* â€” Quick start for minimal faeOS image - *JLCPCB SMT Assembly* â€” â,¬0 setup, â,¬8/stencil, parts at LCSC prices (BOM cost estimator) ### Lab (1h) - **Create bring-up checklist** â†’ `week1/day5\_bringup/checklist.md` - **Git repo init**: `rk3588-carrier` with `.gitignore` for KiCad, `docs/`, `hw/`, `sw/` - **Schematic review checklist** (10 items): Power, DDR, Clocks, Reset, Straps, Debug, Connectors, Mounting, Silkscreen, DRC - **Order plan**: 4-layer 1.6mm FR-4, ENIG, impedance control â€” quote from JLCPCB/PCBWay/AISLER ### Outdoor/Study (0.5h) - Read: *Bringing Up a Custom Board* â€” [Bunnie Huang blog](#) (free) - Join: `##hardware` on Libera.Chat, [r/PrintedCircuitBoard](#), [EEVBlog forum](#) - Plan: Week 2 â€” Schematic completion, BOM optimization, first prototype order --- ## Week 1 Deliverables | Item | Path | Due | |-----|-----|-----| | Spice simulations (diode, MOSFET, inverter, buck) | `week1/day1\_spice/` | Day 1 | | KiCad project: power + RK3588 + DDR4 start | `week1/day2\_kicad/rk3588\_carrier.kicad\_pro` | Day 2 | | Buck converter design notes + sim | `week1/day3\_power/buck\_design.md` | Day 3 | | DDR4 routing constraints + eye sim | `week1/day4\_ddr/ddr\_constraints.md` | Day 4 | | Bring-up checklist + git repo | `week1/day5\_bringup/` | Day 5 | | **Week 1 Reflection** (1 page) | `week1/reflection.md` | End of week | --- ## Free Resource Index (Bookmark These) | Category | Resource | Access | |-----|-----|-----| | Semiconductor Physics | MIT 6.007 OCW | ocw.mit.edu | | MOSFET/Analog | All About Circuits Vol. 3 | allaboutcircuits.com/textbook | | PCB Design | KiCad 8 Tutorial | docs.kicad.org/8.0 | | IPC Standards | Sierra Circuits Whitepapers | sierraproto.com/resources | | High-Speed Design | Rick Hartley Notes | [si-list archives / YouTube](#) | | Power Electronics | TI Power Supply Seminar | ti.com/training | | DDR4 | Micron TN-47-01 | micron.com/support | | Signal Integrity | Eric Bogatin YouTube | youtube.com/@EricBogatin | | RK3588 Docs | Rockchip GitHub | github.com/rockchip-linux | | ARM Firmware | Trusted Firmware-A | trustedfirmware.org | | Manufacturing | JLCPCB/PCBWay Capabilities | jlcpcb.com/capability | | Community | Libera.Chat [##hardware](#) | irc.libera.chat | --- ## Budget Tracker (Week 1) | Item | Cost | Status | |-----|-----|-----| | USB Microscope (500x, 1080p) | â,¬50 | Pending | | TS100 Soldering Iron + Tips | â,¬60 | Pending | | Digital Multimeter (UNI-T UT61E+) | â,¬30 | Pending | | SMD Practice Kit (0201/0402/BGA) | â,¬50 | Pending | | **Subtotal (Month 1-3)** | **â,¬190** | â€” | | Hot-Air Rework (shared/makerspace) | â,¬0 | Access arranged | | 4-Ch Oscilloscope (shared/makerspace) | â,¬0 | Access arranged | | **Total New Spend** | **â,¬190** | **Within â,¬1,000** | --- *Generated: 2026-08-06 | Next: Week 2 â€” Schematic Completion & BOM*